

$$\text{rel. freq} = \frac{\text{freq}}{n}$$

$$k = \sqrt{n}$$

$$w = \frac{\text{max} - \text{min}}{k}$$

$$\text{density} = \frac{\text{rel. freq}}{w}$$

$$\text{mean} : \bar{x} = \frac{\text{sum of all obs.}}{n}$$

$$P_k(\text{prcntl}) = x_{\frac{kn}{100}}$$

$$\text{IQR} = Q_3 - Q_1$$

$$\text{Var} : s^2 = \frac{1}{n-1} \left\{ \sum_{i=1}^n x_i^2 - n\bar{x} \right\}$$

$$\text{SD} : s = \sqrt{s^2}$$

$$P(E) = \frac{\# E \text{ occurred}}{\# S \text{ occurred}}$$

(given all outcomes in S are equally likely)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\text{Disjoint} : P(A \cap B) = 0$$

$$\text{Indpndnt} : P(A \cap B) = P(A)P(B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A|B)P(B) \vee P(B|A)P(A)$$

$$P(A) = P(A|B)P(B) + P(A|\bar{B})P(\bar{B})$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(A|B)P(B)}{P(A)}$$